Kidney Function and Survival in Heart Failure

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Question

Does reduced kidney function (eGFR) predict death in heart failure patients, independently of age, ejection fraction and other clinical factors?

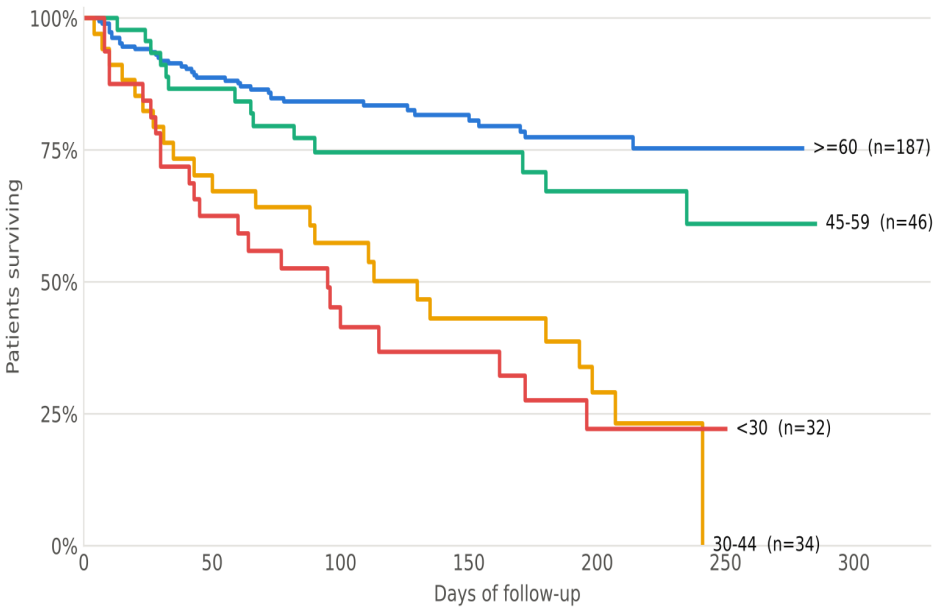
Data and method

- **Data:** clinical records of 299 heart failure patients (NYHA III–IV) from a cardiology hospital in Faisalabad, Pakistan, 2015; 96 deaths over 4–285 days of follow-up (UCI dataset 519, CC BY 4.0).
- **Exposure:** eGFR derived from serum creatinine, age and sex with the race-free CKD-EPI 2021 equation; grouped at standard CKD cut-points.
- **Analysis (R):** mortality rates per 100 person-years, Kaplan–Meier curves and log-rank test; Cox proportional hazards models adjusted for 8 clinical factors; proportional-hazards and linearity checks; two sensitivity analyses.

Key findings

eGFR (mL/min/1.73m^2)	Patients	Deaths	Per 100 person-years	Adjusted HR (95% CI)
60 or above	187	37	52	1 (reference)
45–59	46	14	80	1.38 (0.73–2.61)
30–44	34	23	228	2.58 (1.44–4.63)
Below 30	32	22	268	3.85 (2.12–7.00)

Survival falls sharply once eGFR is below 45 mL/min/1.73m²



Kaplan–Meier estimates by eGFR group (CKD-EPI 2021). 299 heart failure patients, 96 deaths. The 30–44 line drops to 0% when its last patient at risk dies – a small-numbers effect.

- **Each 10 mL/min/1.73m^2 lower eGFR was linked to a 22% higher hazard of death** (HR 1.22, 1.11–1.34) after adjustment.
- The risk was concentrated below eGFR 60 (HR 1.32 per 10 lower) and not clear above it.
- Results held after excluding 19 patients with likely imputed creatinine values (HR 1.21) and when using creatinine instead of eGFR.

Data quality and limitations

Some values equal the column mean (e.g. 25 platelet counts), suggesting they were filled in before release; these were flagged and tested. The sample is small and single-centre, follow-up is short, eGFR comes from one creatinine reading, and the design shows associations, not causes.

Skills shown

Clinical data checking · derived clinical variables (eGFR, CKD stages) · survival analysis (Kaplan–Meier, log-rank, Cox) · model checking and sensitivity analysis · reproducible R code · clear reporting of limitations